

AM 20 Week 1 Section Solutions

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Problem 1: The Attendance Problem

Note that this problem can be solved via the separation of variables method or the method of integrating factor, so I will solve both here. This problem can be divided into two sections, solving the ODE and then the initial value problem. We have,

Separation of Variables

$$\begin{aligned}\frac{dp}{dt} &= \frac{1}{2}p - 100 \\ \frac{dp}{p-200} &= \frac{dt}{2} \\ \int \frac{dp}{p-200} &= \int \frac{dt}{2} \\ \ln |p-200| &= \frac{t}{2} + C \\ p-200 &= e^{t/2+C} \\ p &= 200 + Ce^{t/2}\end{aligned}$$

Method of integrating Factor

$$\begin{aligned}\frac{dp}{dt} &= \frac{1}{2}p - 100 \\ p' - \frac{1}{2}p &= -100 \\ \mu(x) &= e^{-\int 1/2 dt} \\ \mu(x) &= e^{-t/2} \\ \int e^{-t/2} \left(p' - \frac{1}{2}p \right) dt &= \int -100e^{-t/2} dt \\ e^{-t/2}p &= 200e^{-t/2} + C \\ p &= 200 + Ce^{t/2}\end{aligned}$$

Now that we have a general solution for p we can solve for the IVP by taking in the initial condition $p(0) = p_0$. We have,

$$\begin{aligned}p(0) &= 200 + Ce^0 = p_0 \\ C &= p_0 - 200 \\ p(t) &= 200 + (p_0 - 200)e^{t/2}\end{aligned}$$

In order to solve for parts a and b, we will now solve for p_0 when we have $p(3) = s$. Notice that to solve for part a, we use $s = 150$ and for part b we use $s = 0$. The solution will hold for any general s . We have,

$$\begin{aligned}p(3) &= 200 + (p_0 - 200)e^{3/2} = s \\ p_0 - 200 &= (s - 200)e^{-3/2} \\ p_0 &= 200 + (s - 200)e^{-3/2}\end{aligned}$$

Finally, we have part c, where we find the steady state of the ODE. Notice that if we have $p_0 = 200$ that the attendance becomes stationary. This can be observed in one of two ways. Most intuitively, we look at the original differential equation. We have,

$$\begin{aligned}\left. \frac{dp}{dt} \right|_{p_0=200} &= \frac{1}{2}(200) - 100 \\ \left. \frac{dp}{dt} \right|_{p_0=200} &= 100 - 100 = 0.\end{aligned}$$

Therefore, when the attendance equals 200, the rate of change of the attendance is zero and remains at 200 for all future time. The other way of seeing this is going to our general solution. If $p_0 = 200$ we have,

$$\begin{aligned} p(t) &= 200 + (200 - 200)e^{t/2} \\ p(t) &= 200 \end{aligned}$$

This indicates that our IVP becomes time independent if our initial condition is $p_0 = 200$, this matches our analysis of the original ODE.

Problem 2: The Falling Object Problem

Here we are given the falling object equation with linear drag,

$$m \frac{dv}{dt} = mg - \gamma v, \quad v(0) = 0,$$

where m , g , and γ are positive constants and v is positive downward. This problem can be divided into three sections: checking that the proposed solution actually solves the IVP, finding the time to reach 90 percent of the terminal velocity, and then plugging in the numbers from lecture.

For part a, we are not asked to solve the ODE, only to check that

$$v(t) = \frac{mg}{\gamma} (1 - e^{-\gamma t/m})$$

is a solution. To do this we compute the left-hand side and the right-hand side of the ODE separately and compare them. We have,

Left-Hand Side

Right-Hand Side

$$\begin{aligned} v(t) &= \frac{mg}{\gamma} - \frac{mg}{\gamma} e^{-\gamma t/m} & mg - \gamma v &= mg - \gamma \frac{mg}{\gamma} (1 - e^{-\gamma t/m}) \\ v'(t) &= -\frac{mg}{\gamma} \left(-\frac{\gamma}{m}\right) e^{-\gamma t/m} & mg - \gamma v &= mg - mg (1 - e^{-\gamma t/m}) \\ v'(t) &= g e^{-\gamma t/m} & mg - \gamma v &= mg - mg + mg e^{-\gamma t/m} \\ mv'(t) &= m g e^{-\gamma t/m} & mg - \gamma v &= mg e^{-\gamma t/m} \end{aligned}$$

Both sides are equal to $mg e^{-\gamma t/m}$ for every t , so $v(t)$ satisfies the ODE. We still need to check the initial condition,

$$v(0) = \frac{mg}{\gamma} (1 - e^0) = \frac{mg}{\gamma} (1 - 1) = 0.$$

Therefore $v(t)$ solves the IVP. Note that we could also have found this solution directly using either of the methods from the attendance problem.

For part b, we first need the terminal velocity. Just like the steady state in the attendance problem, the terminal velocity is the velocity at which the rate of change of the velocity is zero. We have,

$$\begin{aligned} m \frac{dv}{dt} \Big|_{v=v_T} &= mg - \gamma v_T = 0 \\ v_T &= \frac{mg}{\gamma}. \end{aligned}$$

We can also see this from our solution, since $e^{-\gamma t/m} \rightarrow 0$ as $t \rightarrow \infty$ and so $v(t) \rightarrow mg/\gamma$. Now, rather than only solving for 90 percent, we will solve for the time t_f at which the object reaches a general fraction f of

the terminal velocity, where $0 < f < 1$. Notice that to solve for part b we use $f = 0.9$. We have,

$$\begin{aligned}v(t_f) &= f v_T \\ \frac{mg}{\gamma} \left(1 - e^{-\gamma t_f/m}\right) &= f \frac{mg}{\gamma} \\ 1 - e^{-\gamma t_f/m} &= f \\ e^{-\gamma t_f/m} &= 1 - f \\ -\frac{\gamma t_f}{m} &= \ln(1 - f) \\ t_f &= -\frac{m}{\gamma} \ln(1 - f) = \frac{m}{\gamma} \ln\left(\frac{1}{1 - f}\right)\end{aligned}$$

Plugging in $f = 0.9$ gives $1/(1 - f) = 10$, so

$$t_{90} = \frac{m}{\gamma} \ln 10.$$

Notice that t_{90} does not depend on g at all. If we doubled g , we would double the terminal velocity, but the object would take exactly the same amount of time to reach 90 percent of it. The ratio m/γ is the time scale of the problem.

Finally, we have part c, where we use the lecture object with $m = 10$ kg and $\gamma = 2$ kg/sec, so that $m/\gamma = 5$ sec and $v_T = (10)(9.8)/2 = 49$ m/sec. We have,

$$t_{90} = 5 \ln 10 \approx 11.51 \text{ sec.}$$

To decide whether the object gets there before it lands, we need the time it takes to fall 300 meters. Let $x(t)$ be the distance fallen, so that $dx/dt = v(t)$ and $x(0) = 0$. Integrating our solution,

$$\begin{aligned}x(t) &= \int_0^t 49 \left(1 - e^{-s/5}\right) ds \\ x(t) &= 49t - 245 \left(1 - e^{-t/5}\right)\end{aligned}$$

Setting $x(t_g) = 300$ gives an equation that mixes a linear and an exponential term, so we solve it numerically (by trial and error, bisection, or a calculator),

$$\begin{aligned}49t_g - 245 \left(1 - e^{-t_g/5}\right) &= 300 \\ t_g &\approx 10.51 \text{ sec.}\end{aligned}$$

Since $t_g \approx 10.51 < 11.51 \approx t_{90}$, the object lands about one second before it would reach 90 percent of its terminal velocity, so the answer is no. We can double check this with the impact speed,

$$v(t_g) = 49 \left(1 - e^{-10.51/5}\right) \approx 43.01 \text{ m/sec,}$$

which is only $43.01/49 \approx 0.878$, or about 88 percent, of the terminal velocity.

Problem 3: Classifying ODEs

Here we are asked about the three equations

$$(a) \quad y'' + t^n y = 0, \quad (b) \quad y'' + y^n = 0, \quad (c) \quad y'' + (y')^2 = 0,$$

where n is a constant. Before starting, recall what it means for an ODE to be linear. A linear ODE of order k can be written in the form

$$a_k(t)y^{(k)} + \cdots + a_1(t)y' + a_0(t)y = g(t).$$

That is, y and its derivatives only appear to the first power, they are never inside another function (like $\sin$ or $e^{(\cdot)}$), and they are never multiplied together. The coefficients $a_i(t)$ and the forcing $g(t)$, on the other hand, can depend on the independent variable t in any way they like.

For part a, notice that the exponent n sits on the independent variable t , not on the unknown y . Comparing with the linear form, we have,

$$y'' + t^n y = 0$$

$$a_2(t) = 1, \quad a_1(t) = 0, \quad a_0(t) = t^n, \quad g(t) = 0$$

Since t^n is just a coefficient which depends on t , the equation fits the linear form no matter what n is. Therefore equation (a) is linear for every value of n .

For part b, the exponent now sits on the unknown y , so the value of n matters. Let's check the cases,

$$\begin{array}{ll} y'' + y^0 = 0 & y'' + y^1 = 0 \\ y'' + 1 = 0 & y'' + y = 0 \\ y'' = -1 & a_2 = 1, \quad a_0 = 1, \quad g = 0 \end{array}$$

For $n = 0$ the unknown only appears as y'' to the first power, and the constant -1 is a forcing term $g(t) = -1$, so the equation is linear. For $n = 1$ the equation is clearly linear. For any other value of n , the term y^n raises the unknown to a power other than one, so the equation is nonlinear. Therefore equation (b) is linear only for $n = 0$ and $n = 1$.

Notice that parts a and b look almost identical. The only difference is whether the power is on the independent variable or the dependent variable, which is why it is always worth identifying the variables before testing linearity.

Finally, we have part c, where we go through all four classifications one at a time for $y'' + (y')^2 = 0$.

- **ODE or PDE:** The primes denote derivatives with respect to t , and there is no other variable we differentiate with respect to. One independent variable means this is an ODE.
- **Scalar or system:** The only unknown function is y , so this is a scalar equation.
- **Order:** The highest derivative is y'' , so the equation is second order. Note that the square on y' does not change the order, since squaring a first derivative does not produce a second derivative.
- **Linear or nonlinear:** The term $(y')^2$ raises a derivative of the unknown to the second power, so the equation is nonlinear.

As a bonus, neither term contains t explicitly, so the equation is also time-invariant (autonomous).